



UNIVERSITY OF TORONTO
FACULTY OF APPLIED SCIENCE & ENGINEERING

APS110/164

Engineering Chemistry and Materials Science

Tutorial 8: Optical Properties and Structure of Matter

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Course Announcements

1. Midterm:

Grades: are available now

Midterm is out of 34, mark on Quercus is out of 30

If your grade input is incorrect, please let me know

1. Quiz 3:

Friday, November 25

Coverage: Course schedule (on the homepage)

2. Problem Set 3:

Posting: Early/mid this week

Due: Dec 2

Contents Covered in this Tutorial

	9. Quick! We need a better model!	Structure of Matter	The Bohr model	Use the Bohr model of the atom to explain absorption and emission of energy accompanying transitions in electron energy levels
	9. Quick! We need a better model!	Electromagnetic Radiation	Fluorescent lamps, highlighters, and streetlamps	Identify the shortcomings of the Bohr model, specifically the need for four quantum numbers to fully describe the energy level of an electron
	9. Quick! We need a better model!	Structure of Matter	Electronic configuration, the four quantum numbers - something intuitive	Provide intuitive descriptions of each of the four quantum numbers
November 14, 2022	9. Quick! We need a better model!	Structure of Matter	The four quantum numbers - allowable values	Determine the electron configuration of a neutral atom or charged ion Identify the allowable values for each of the quantum numbers
	9. Quick! We need a better model!	Bonding	Electronic configuration, Octet Stability	Apply the principle of octet stability to explain covalent, ionic and metallic (sea-of-electrons) bonding Differentiate between the mechanism of formation of each of the primary bonds and provide examples of several properties that can be explained by each

	9. Quick! We need a better model!	Bonding	The Covalent Bond	Apply the principle of octet stability to explain covalent, ionic and metallic (sea-of-electrons) bonding Differentiate between the mechanism of formation of each of the primary bonds and provide examples of several properties that can be explained by each
	9. Quick! We need a better model!	Bonding	The Ionic Bond	Apply the principle of octet stability to explain covalent, ionic and metallic (sea-of-electrons) bonding Differentiate between the mechanism of formation of each of the primary bonds and provide examples of several properties that can be explained by each
	9. Quick! We need a better model!	Bonding	The Metallic Bond	Apply the principle of octet stability to explain covalent, ionic and metallic (sea-of-electrons) bonding Differentiate between the mechanism of formation of each of the primary bonds and provide examples of several properties that can be explained by each
November 21, 2022	9. Quick! We need a better model!	Bonding	Ordered solids revisited: achieving low energy	Conclude that crystals form in ordered solids according to a decrease in energy
	9. Quick! We need a better model!	The Band Theory of Solids	The band theory of solids	Demonstrate how the allowable energy states in an isolated atom separate into many closely spaced states in the formation of a solid leading to a band structure Apply the band theory of solids to explain the classification of materials as metals, insulators, and semiconductors

	9. Quick! We need a better model!	The Band Theory of Solids	Extrinsic and Intrinsic semiconductors	Apply the band theory of solids to explain the classification of materials as metals, insulators, and semiconductors
	9. Quick! We need a better model!	The Band Theory of Solids	Extrinsic and Intrinsic semiconductors	- Given a semiconductor material and a dopant, determine if a p-type or n-type semiconductor is created - Given the electron and hole mobilities and concentrations, calculate the conductivity of a semiconductor

Content Review

Fundamental Concepts – The Atom:

Electron charge = -1.602×10^{-19} [C]

Proton charge = $+1.602 \times 10^{-19}$ [C]

Neutron charge = 0

Proton mass = 1.67×10^{-27} [kg]

Neutron mass = 1.67×10^{-27} [kg]

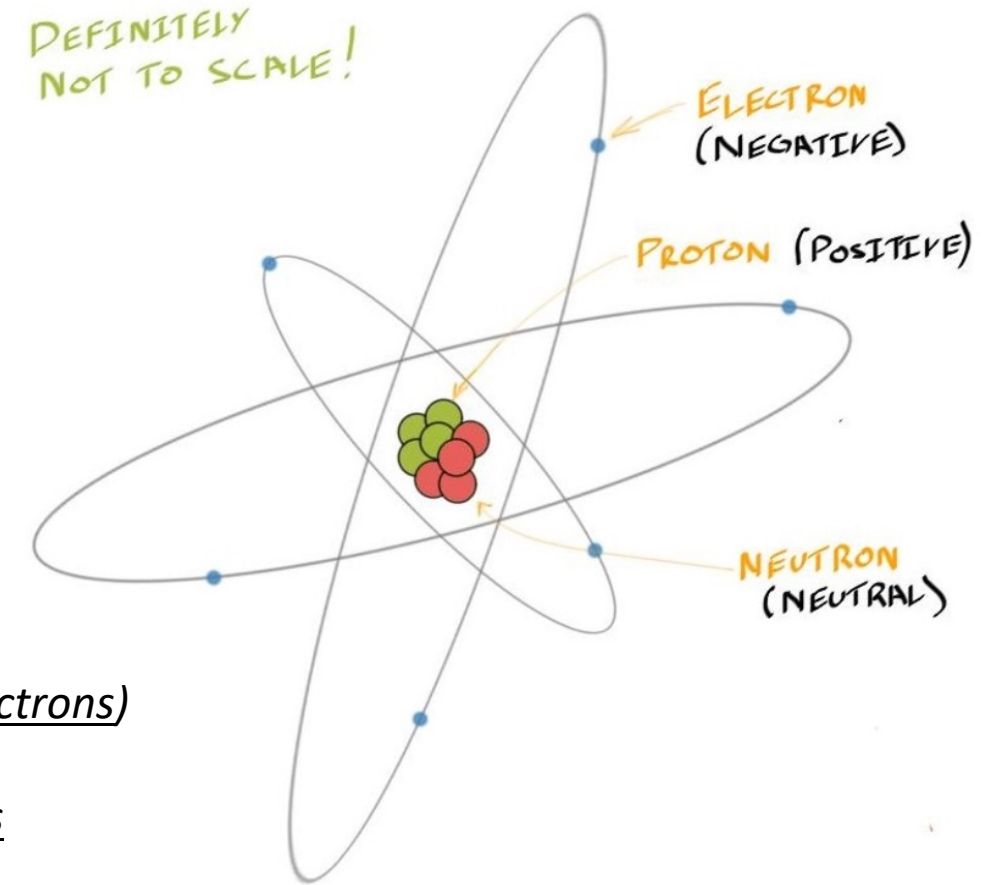
Electron mass = 9.11×10^{-31} [kg]

Z [atomic number]: number of protons (if neutral, equals the number of electrons)

A [atomic mass] eq $\approx$ mass of nucleus = mass of protons + mass of neutrons

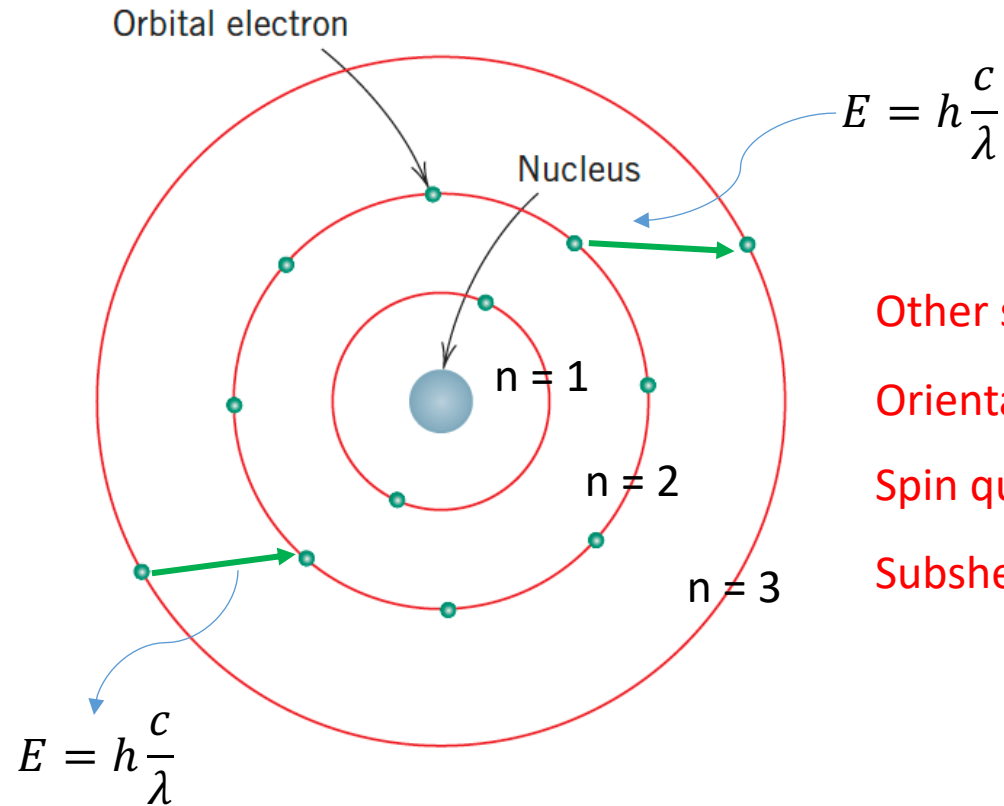
Isotopes of an element: atoms of an element with different numbers of neutrons (and, thereby, different atomic weights)

Ex: ^{14}C and ^{12}C !



Content Review

The Bohr Model of the Atom:



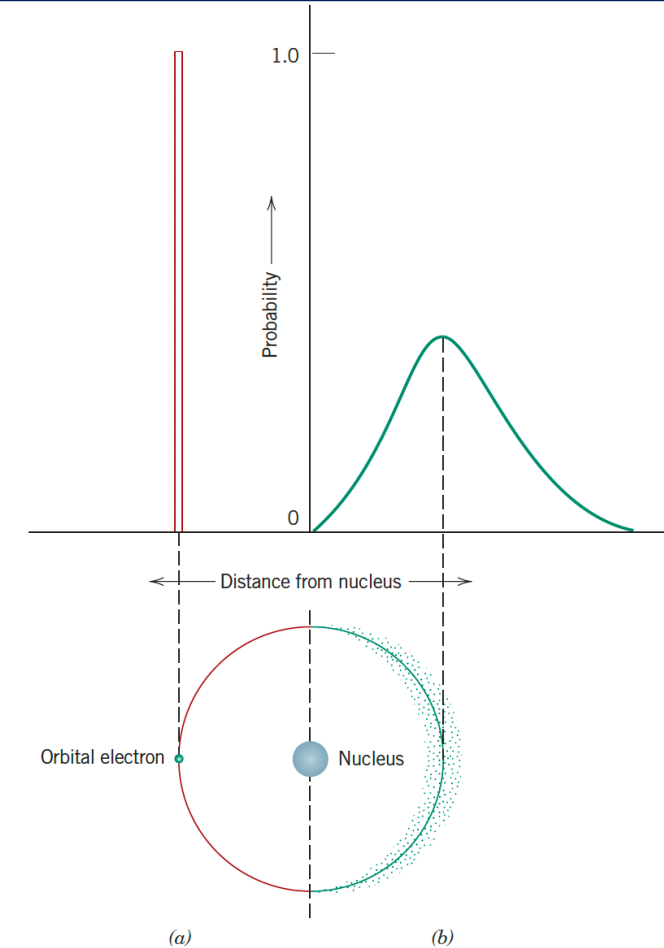
Other shapes of orbitals?

Orientations of orbitals?

Spin quantum number?

Subshells/distribution for energy?

Wave-Mechanical Model of the Atom:



Comparison of the (a) Bohr and (b) wave-mechanical atom models in terms of electron distribution.

Content Review

Four Quantum Numbers:

Principal Quantum number: n

Refers to: the **size** of the electron orbit (shell)

$$n = 1, 2, 3, 4, 5, \dots$$

$$n = K, L, M, N, O, \dots$$

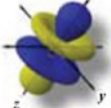
Angular Momentum/Azimuthal Quantum Number: l

Refers to: the **shape** of the electron orbital (subshell)

$$l = 0, 1, 2, 3, 4, \dots, n-1$$

$$l = s, p, d, f$$

Courtesy: study.com

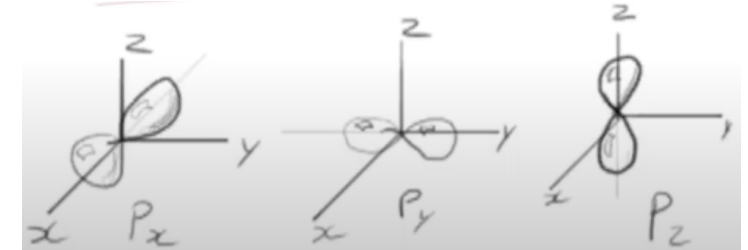
Angular Momentum Quantum Number, ℓ	Name of Subshell	Shape	
0	s	Sphere	
1	p	Dumbbell	
2	d	Complex/double dumbbell	
3	f	More complex/multiple lobes	

Magnetic Quantum Number: m_l

Refers to: the **spatial orientation** of the electron orbital

$$-l \leq m_l \leq +l$$

$$\# \text{ of orbitals} = 2(l) + 1$$

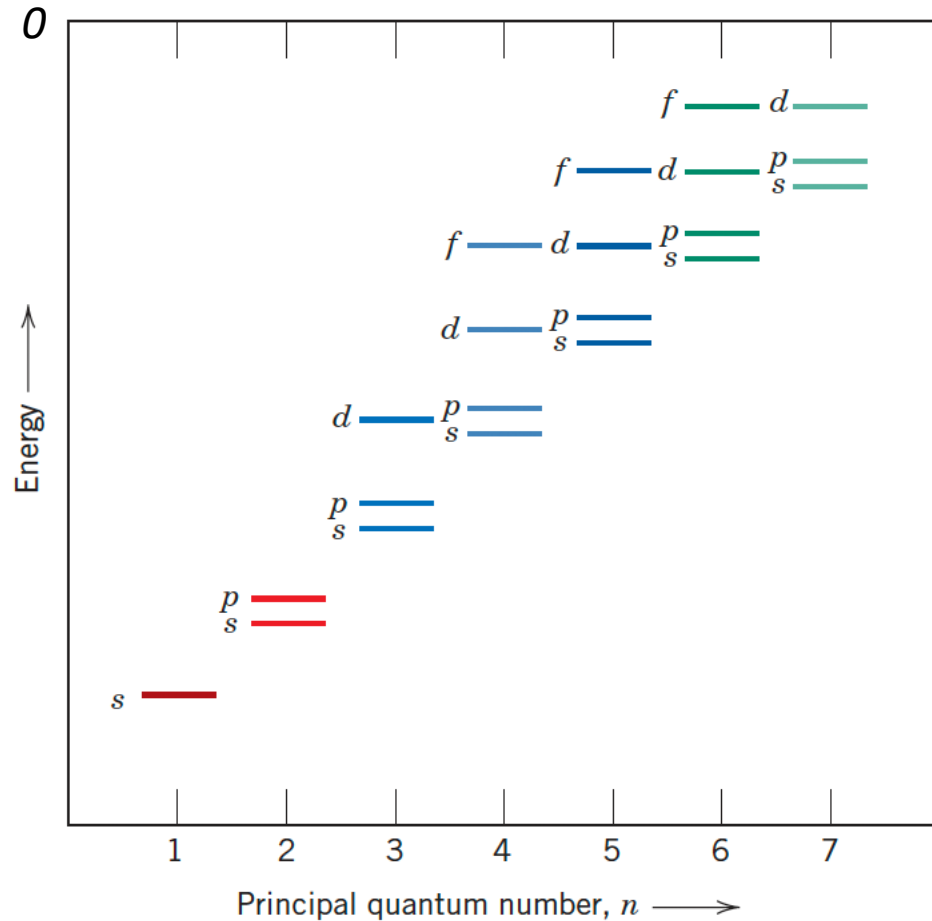


Spin Quantum Number: m_s

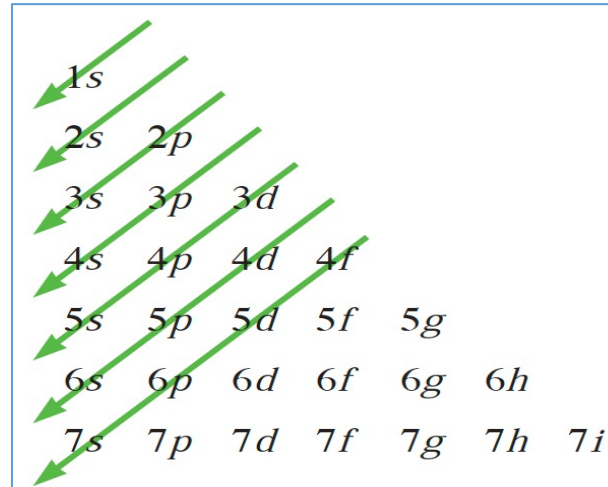
$$m_s = \pm \frac{1}{2}$$

Content Review

Electron Configurations:



The Aufbau Principle:

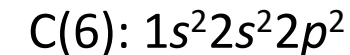


Pauli exclusion principle:

Each electron state (orbital) can hold no more than two electrons, which must have opposite spins.

Thus, s, p, d, and f subshells may each accommodate, respectively, a total of 2, 6, 10, and 14 electrons

Notation:



Content Review

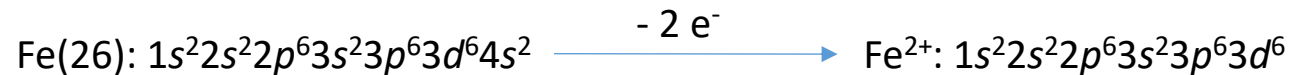
Electron Configurations: Points/Exceptions:

1. **Abbreviate Notation:** using the electron configuration of inert gases in square brackets

<i>Element</i>	<i>Symbol</i>	<i>Atomic Number</i>	<i>Electron Configuration</i>	
Helium	He	2	$1s^2$	C(6): $1s^2 2s^2 2p^2 = [\text{He}] 2s^2 2p^2$
Neon	Ne	10	$1s^2 2s^2 2p^6$	Al(13): $1s^2 2s^2 2p^6 3s^2 3p^1 = [\text{Ne}] 3s^2 3p^1$
Argon	Ar	18	$1s^2 2s^2 2p^6 3s^2 3p^6$	Ca(20): $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 = [\text{Ar}] 4s^2$

Octet Stability: $ns^2 np^6$

2. **Always** write the electron configuration in order of increasing the principal quantum number!

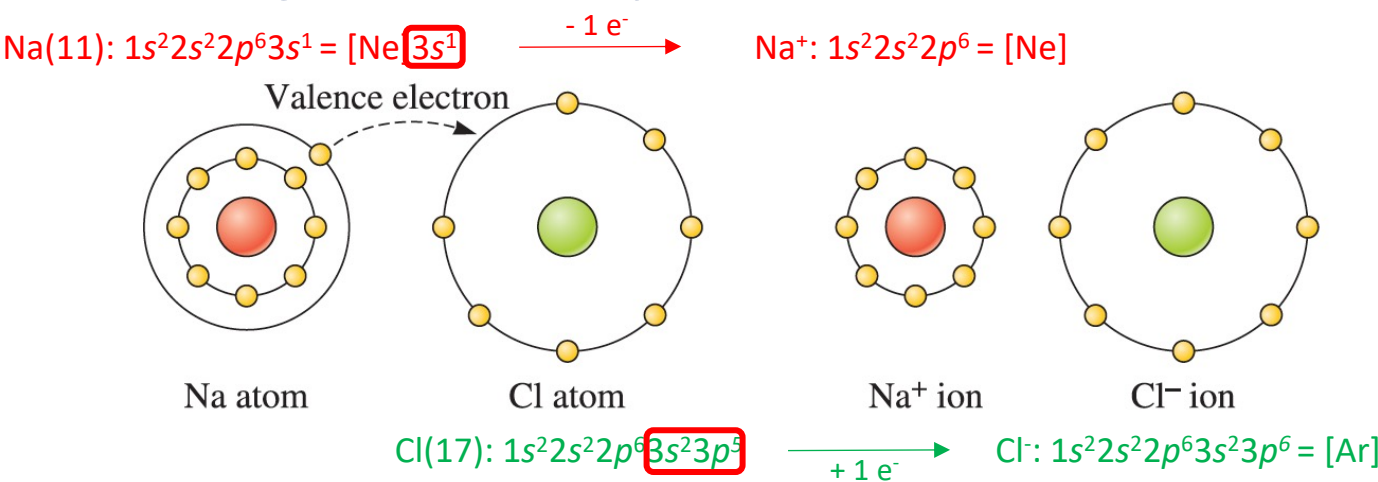


3. The 3d subshell **half-filled (5 electrons)** or **completely-filled (10 electrons)** is slightly lower energy!

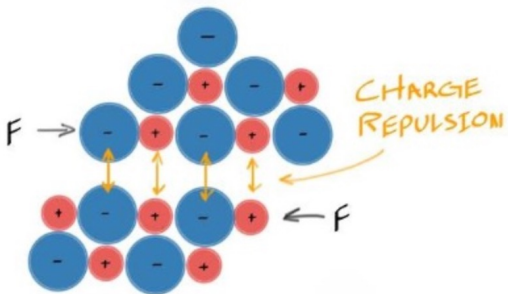
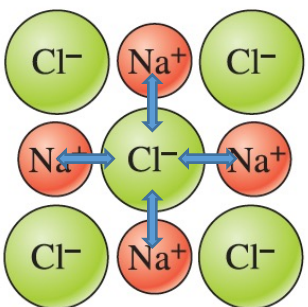


Primary Interatomic Bonding:

Ionic Bonding (electron transfer):



Non-directional!



- ✓ Hard/brittle
- ✓ Electrically/thermally insulative
- ✓ High melting temperatures

➤ Why Does Salt Form a Crystal?

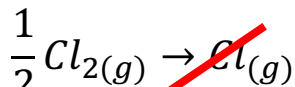
Step 1: sublimation of solid sodium

Step 2: ionization of gas sodium atom

Step 3: bond dissociation of a chlorine molecule

Step 4: chlorine acquiring an electron (electron affinity reaction)

Step 5: forming the solid ionic crystal from the gas phases

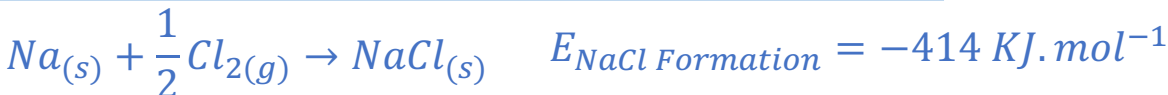


$\Delta H_{\text{sub}} = +109 \text{ KJ} \cdot \text{mol}^{-1}$

$IE_{\text{Na}} = +497 \text{ KJ} \cdot \text{mol}^{-1}$

$BDE_{\text{Cl}_2} = +121 \text{ KJ} \cdot \text{mol}^{-1}$

$EA_{\text{Cl}} = -364 \text{ KJ} \cdot \text{mol}^{-1}$



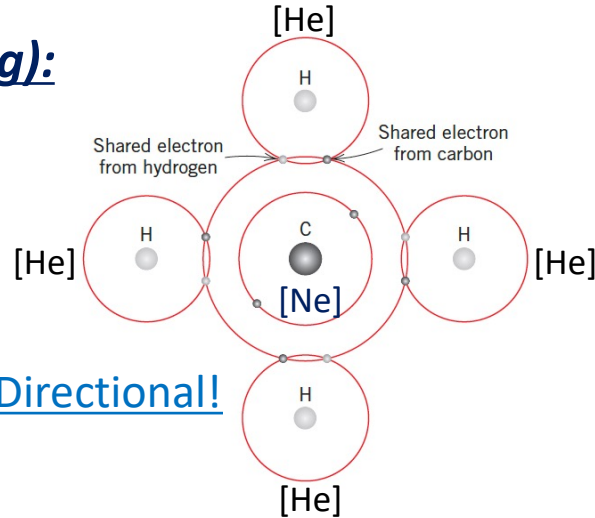
Content Review

Primary Interatomic Bonding:

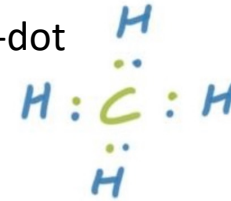
Covalent Bonding (electron sharing):

C(6): $1s^2 2s^2 2p^2$

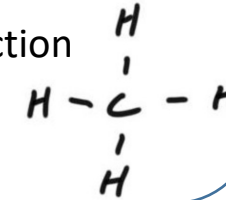
H(1): $1s^1$



❖ Lewis electron-dot diagram



❖ Another depiction

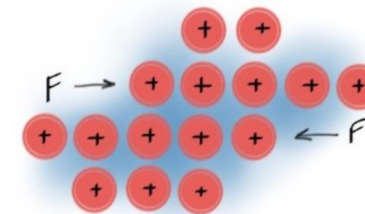
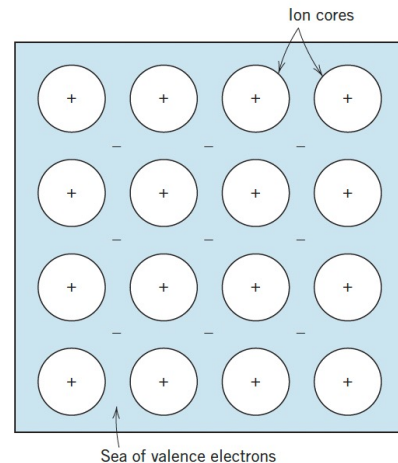


✓ Electrically/thermally insulative

Metallic Bonding:

❖ Sea-of-electrons model:

Non-directional!

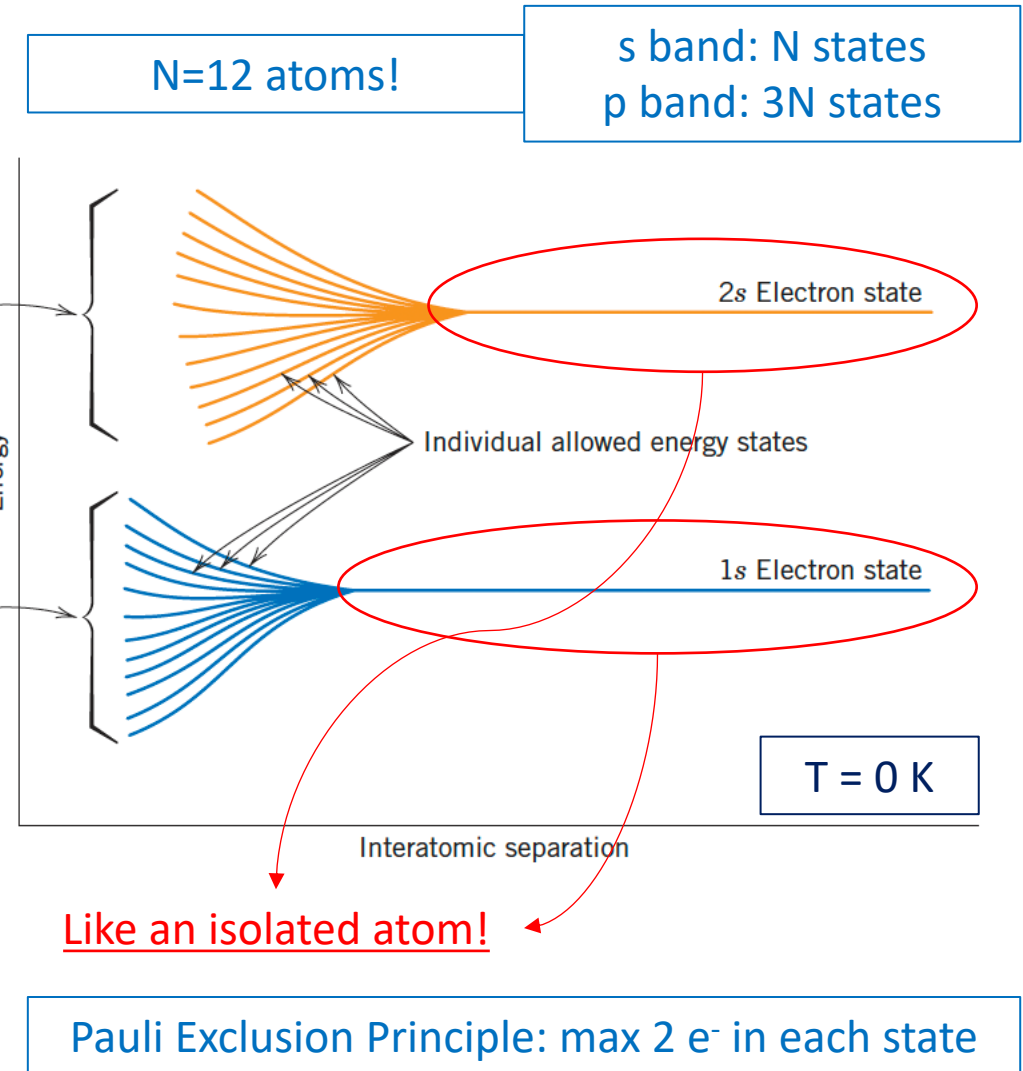
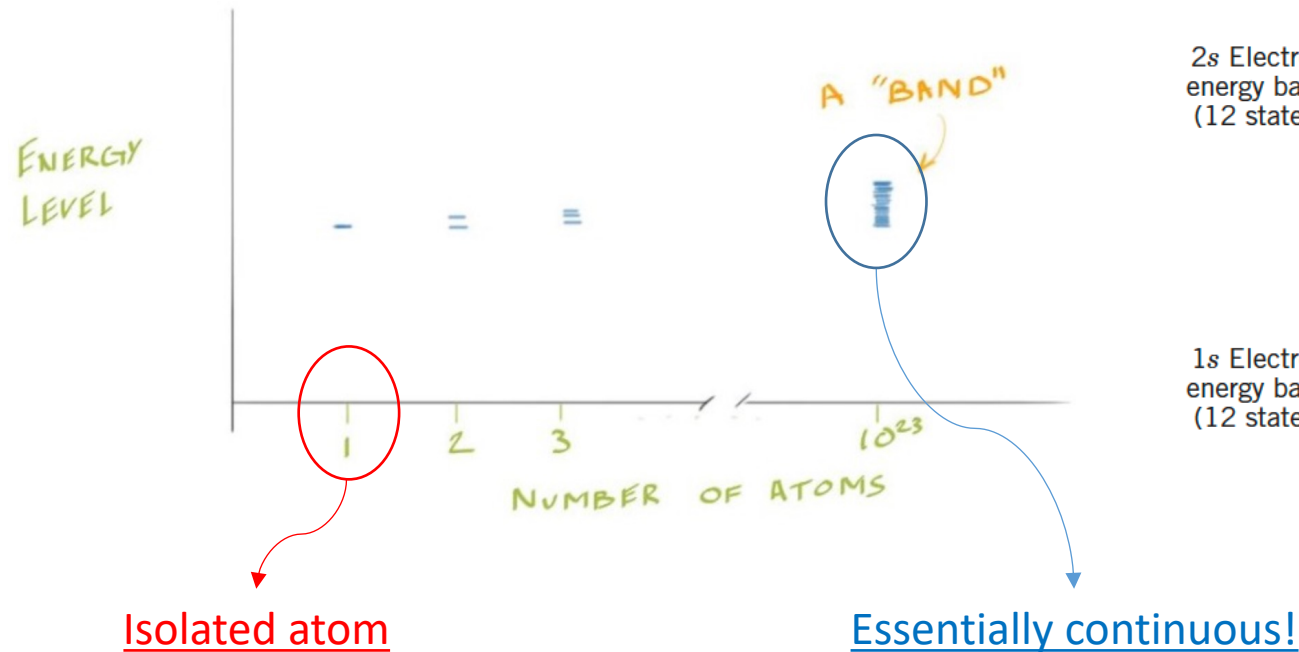


✓ Ductile

✓ Electrically/thermally conductive

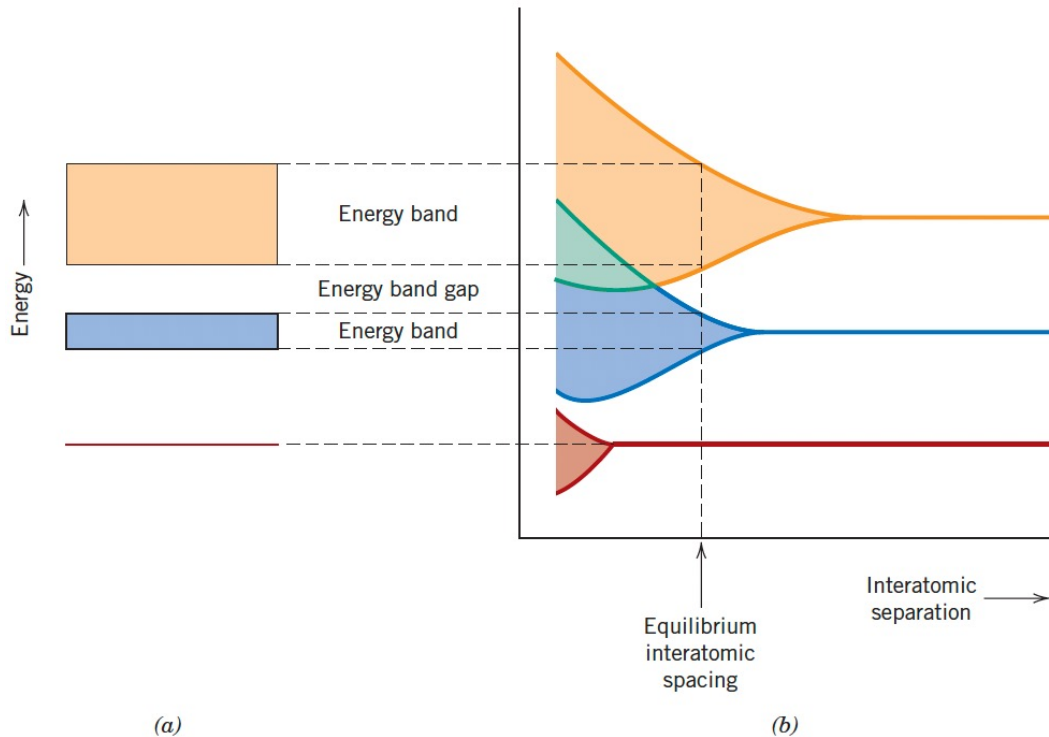
Content Review

Bonding in Solids - The Band Theory:

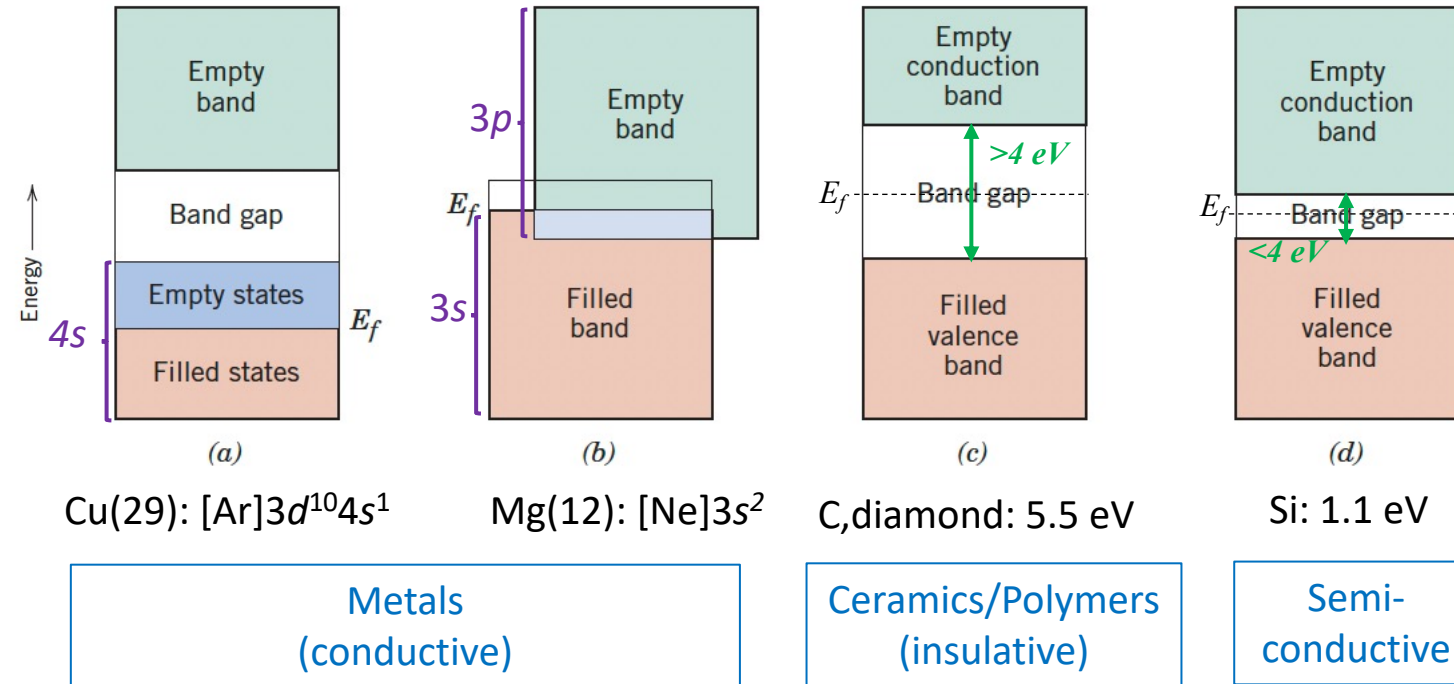


Content Review

Bonding in Solids - The Band Theory:



E_f (fermi level) = The energy corresponding to the highest filled state at 0 K (where the possibility for the presence of an electron is 0.5)



Interaction with visible light (2-3 eV)?

Band gap > 3 eV (eg. SiO₂) → no interactions (optically transparent)

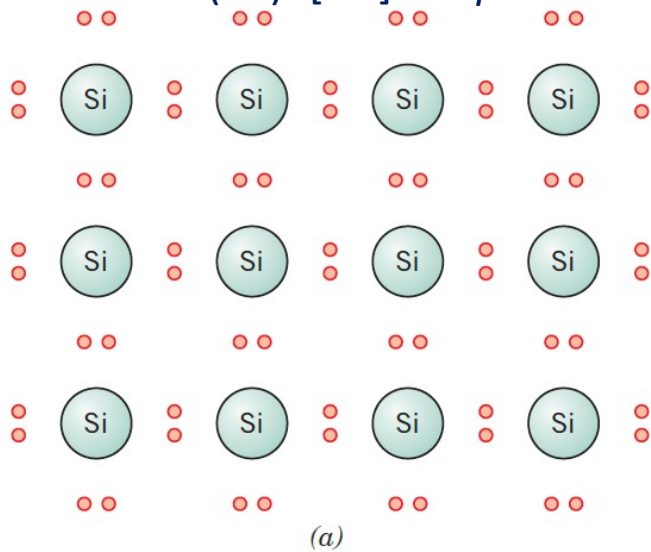
No band gap (metals) → interactions (opaque/shiny)

Content Review

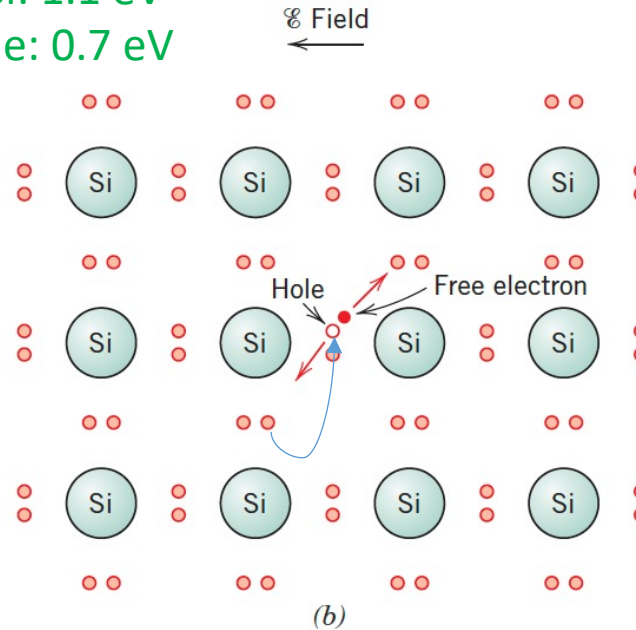
Intrinsic Semiconductors:

Si: 1.1 eV
Ge: 0.7 eV

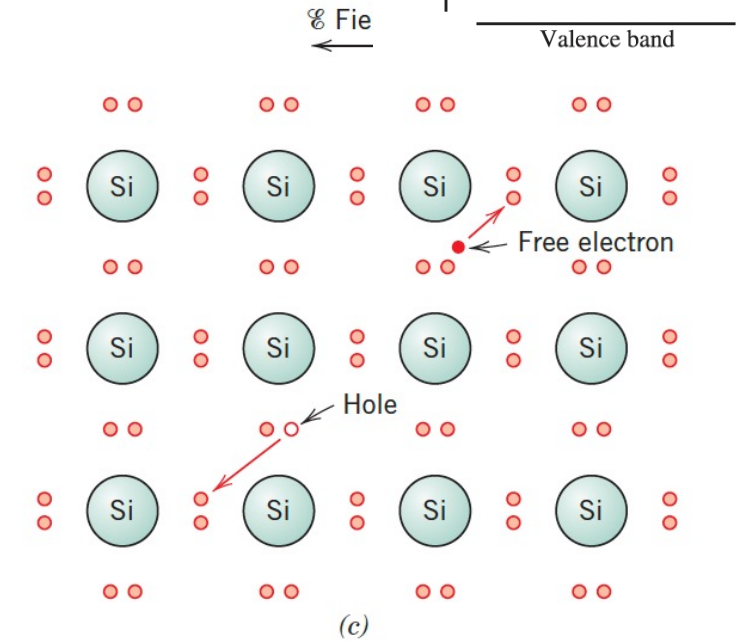
Si(14): [Ne]3s²3p²



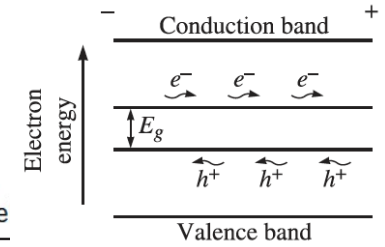
Before excitation ($T = 0$ K, and in the absence of an electric field)



After excitation ($T > 0$ K, or in the presence of an electric field)



Subsequent free-electron and hole motions in response to an external electric field



σ = conductivity [$1/(\Omega \cdot m)$]

n = electron concentration [$\#/m^3$]

p = hole concentration [$\#/m^3$]

q = electron/hole charge = 1.602×10^{-19} [C]

μ_e = electron mobility [$m^2/V \cdot s$]

μ_p = hole mobility [$m^2/V \cdot s$]

$$\sigma = nq\mu_n + pq\mu_p \xrightarrow[n = p]{\text{(one hole is created for every promoted electron)}} \sigma = nq(\mu_n + \mu_p)$$

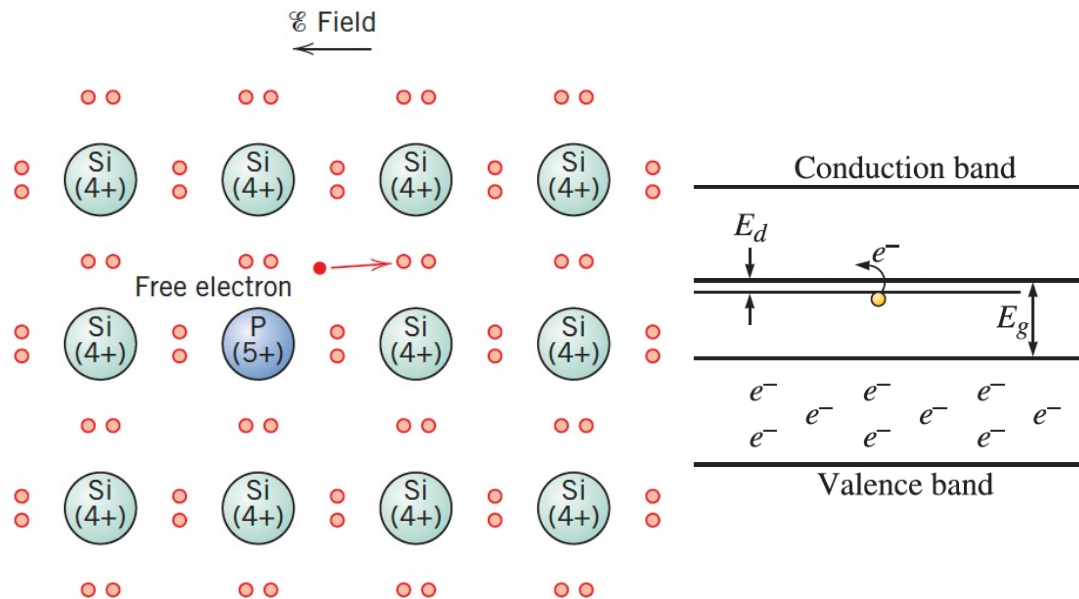
Content Review

Extrinsic Semiconductors:

n-type:

Si(14): [Ne]3s²3p²

P(15): [Ne]3s²3p³

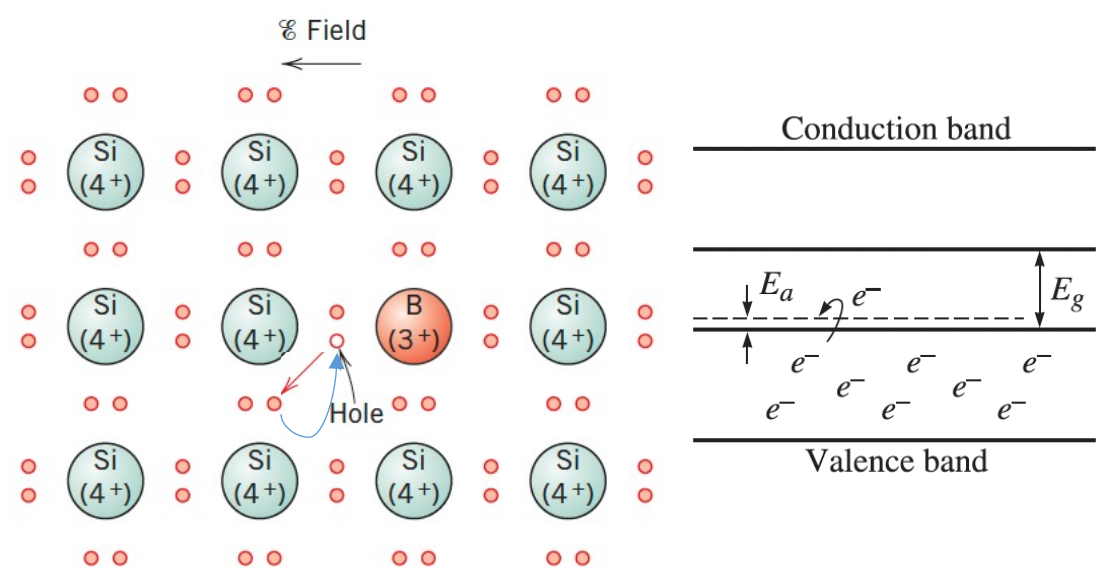


Impurities: from the Group 15 of the periodic table (e.g., P, As, and Sb)

p-type:

Si(14): [Ne]3s²3p²

B(5): [He]2s²2p¹



Impurities: from the Group 13 of the periodic table (e.g., B, Al, and Ga)

$$n \neq p$$

$$\sigma_{n\text{-type}} = nq\mu_n$$

$$\sigma_{p\text{-type}} = pq\mu_p$$

Sample Problems

1. The following questions pertain to the Bohr model and electron configurations:
 - a. What are the two most important concepts for the Bohr model?
 - b. Write the electron configurations for Copper and Chlorine.

Sample Problems

2. At room temperature the electrical conductivity and the electron mobility for aluminum are $3.8 \times 10^7 (\Omega \cdot \text{m})^{-1}$ and $0.0012 \text{ m}^2/\text{V} \cdot \text{s}$, respectively.

(a) Compute the number of free electrons per cubic meter for aluminum at room temperature.

(b) What is the number of free electrons per aluminum atom? Assume a density of 2.7 g/cm^3 .

Tutorial Quiz Questions

1. How many allowable values exist for magnetic quantum number if the angular momentum quantum number is 3.

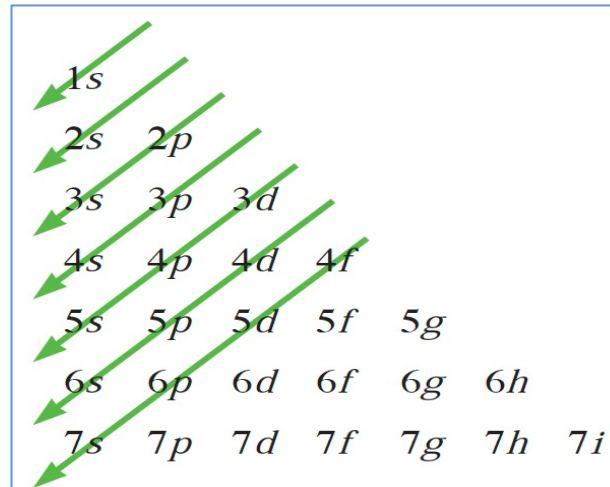
a. 3

b. 4

c. 6

d. 7

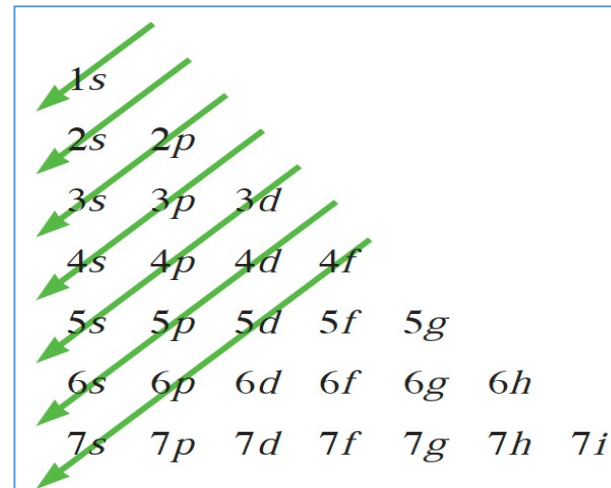
The Aufbau Principle:



Tutorial Quiz Questions

2. What are the values for the quantum numbers for the highest energy electron in ground state sodium ($Z=11$)

The Aufbau Principle:



Tutorial Quiz Questions

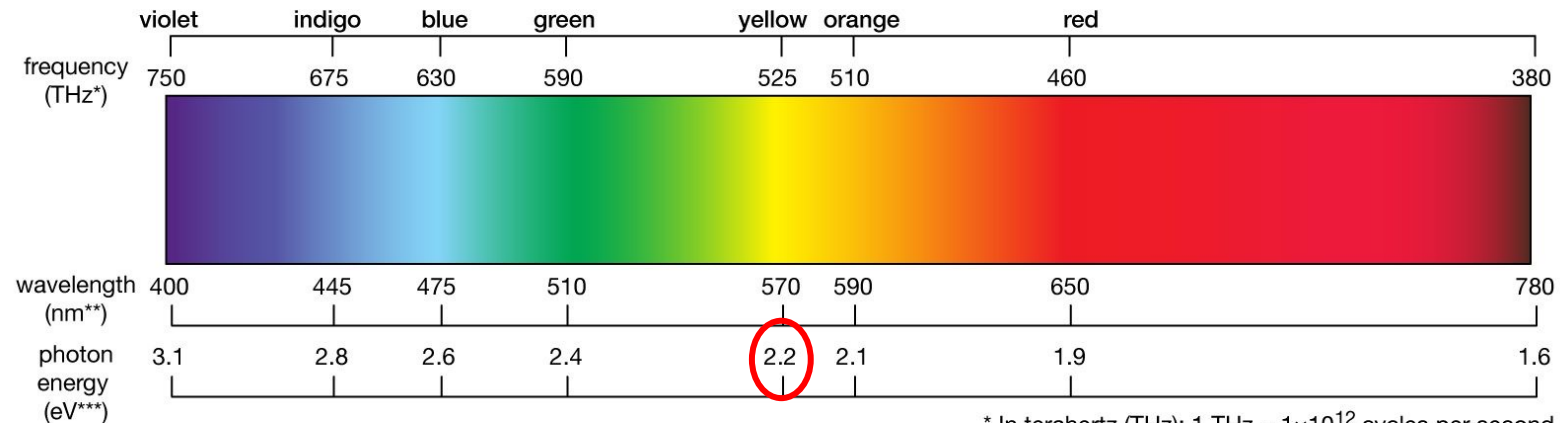
3. Provide the full electron configuration for Nickel²⁺ (Z=28 for ground state Ni)

Tutorial Quiz Questions

4. A non-metallic material has a bandgap of 2.2 eV. In terms of which wavelength of light that passes through the material, which of the following is true?

- a. The material passes all wavelengths of visible light
- b. The material does not pass any wavelengths of visible light (ie: it is opaque)
- c. The material passes red light but not blue light
- d. The material passes blue light but not red light

Light, the visible spectrum



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* In terahertz (THz); 1 THz = 1×10^{12} cycles per second.

** In nanometres (nm); 1 nm = 1×10^{-9} metre.

*** In electron volts (eV).

Tutorial Quiz Questions

5. The following characteristics for both intrinsic and p-type extrinsic GaSb at room temperature. Calculate the hole and electron mobility based on this data --

	σ (Ω m ⁻¹)	n (m ⁻³)	p (m ⁻³)
Intrinsic	2.5×10^{-6}	3.0×10^{13}	3.0×10^{13}
Extrinsic (p-type)	3.6×10^{-5}	4.5×10^{14}	2.0×10^{12}

$$\sigma = nq\mu_n + pq\mu_p$$



Thanks for your attention!
Any Question?

Sample Problem Answers

1.

(1)
a) Electrons are particles moving in discrete orbitals

(2) Electron energy is quantized into shells



(note it is not $4s^2 3d^9$)



Sample Problem Answers

2.

$$n = \frac{\sigma}{|e| \mu_e} = \frac{3.8 \times 10^7}{(1.602 \times 10^{-19})(0.0012 \text{ m}^2/\text{V}\cdot\text{s})} = 1.98 \times 10^{29} \text{ m}^{-3}$$

$$N_{\text{Al}} = \frac{N_A \cdot \rho}{A_{\text{Al}}} = \frac{(6.023 \times 10^{23})(2.71 \text{ g/cm}^3)}{26.98 \text{ g/mol}}$$

$$= 6.03 \times 10^{28} / \text{m}^3$$

of copper atoms
per cubic m.

$$\# \text{ of free electrons} = \frac{n}{N_{\text{Al}}} = \frac{1.98 \times 10^{29} / \text{m}^3}{6.03 \times 10^{28} / \text{m}^3} = 3.28$$

Tutorial Quiz Answers

1. D

2. $n = 3, l = 0, m_l = 0, m_s = -1/2$

3. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^8$

4. C

5.

① Intrinsic $\sigma = n \cdot q \cdot \mu_e + p \cdot q \cdot \mu_h$

$$2.5 \times 10^{-6} = (3 \times 10^3)(1.609 \times 10^{-19}) \mu_e + (3 \times 10^3)(1.609 \times 10^{-19}) \mu_h$$

$$0.52 = \mu_e + \mu_h$$

② Extrinsic

$$3.6 \times 10^5 = (4.5 \times 10^{14})(1.609 \times 10^{-19})(\mu_e) + (2 \times 10^{12})(1.609 \times 10^{-19})(\mu_h)$$

$$112.4 = 225 \mu_e + \mu_h$$

$$\therefore \mu_e = 0.5 \text{ m}^2/\text{V}\cdot\text{s}$$

$$\mu_h = 0.02 \text{ m}^2/\text{V}\cdot\text{s}$$